SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT330

Examination - July Semester 2025

Database Management Systems

Monday, 17 November 2025 01:00 pm - 03:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **08** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A major health company plans to provide telemedicine services by connecting patients to registered health professionals. Health professionals are identified by an id. Their details include a name, contact and specialty. To request for an online consultation, a patient must register with his citizen id, name, date of birth, address, weight and height. When a registered patient makes an online consultation request, the datetime of the request is logged and a request id is generated. The patient needs to provide a few other details regarding the request:

- his availability on the requested date and for the next two days
    - each availability slot is a minimum of 2-hour duration, but it can be indicated with a longer duration. Each slot has a date, a start time and an end time of the patient's availability.
- his medical issues
    - each medication issue is selected from a list of standard medical issues. Only medical issues in the standard list are eligible for online consultation. Each medical issue is identified by a code. It has a name and description. The description helps a patient make a correct selection of medical issue from the standard list.
    - The patient must provide a description of his condition for each selected medical issue.
    - The patient must provide a description of the history for each selected medical issue.

When a match is found, a consultation appointment is fixed with the details: the appointment date and time and an assigned health professional.

Without adding any attribute/identifier not described in the question, **construct** a conceptual entity relationship (ER) model from the statements of requirements to represent the data model, showing the data requirements in **ONE (1)** diagram.

Question 1a

Include entities, with entities name, relevant attributes and identifier.

(10 marks)

Question 1b

Include relationships, with maximum and minimum cardinalities and relationship name.

State your assumption(s) for only data requirements that are not specified.

(10 marks)

Question 2

You are given the table:
Prescription(prescriptionId, deliveryDatetime, address, drugCode, drugName, dosage, instruction)

This table is used by a health company to deliver prescribed drugs after a patient completes an online consultation with a health professional. A prescription is identified by a prescription id. Each prescription contains one or more drugs, each with its required dosage. A drug is identified by a drug code, has a drug name and comes with one or more instructions e.g., after food and no alcohol. Each prescription is assigned a delivery date and time and an address to deliver to.

Question 2a

Formulate and list the functional and multi-valued dependencies (MVD). Do not introduce any new columns not in the given table. State your assumption(s) for only data requirements that are not specified.

(5 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table. **Show composite keys in brackets.**

(3 marks)

Question 2c

Question 2c(i)

Explain whether the MVDs in your answer to Q2(a) are subsumed or not subsumed.

(2 marks)

Question 2c(ii)

Apply the BCNF approach covered in the course text to normalise the table to Boyce Codd Normal Form (BCNF) and Fourth Normal Form (4NF).

- Show how you apply the steps for **ONE (1)** FD **or** MVD.
- For the two tables resulting from the iteration, state whether each of them are in BCNF or in 4NF or otherwise, and **explain** why.

(8 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:
TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:
TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

Note that there is no need to show the derivation steps or explain them for this question part. Simply provide the final tables, the primary keys, the foreign keys and the foreign key constraints.

(7 marks)

Question 3

Question 3a(i)

Transform the ERD in Figure Q3(a)(i) into a logical model.

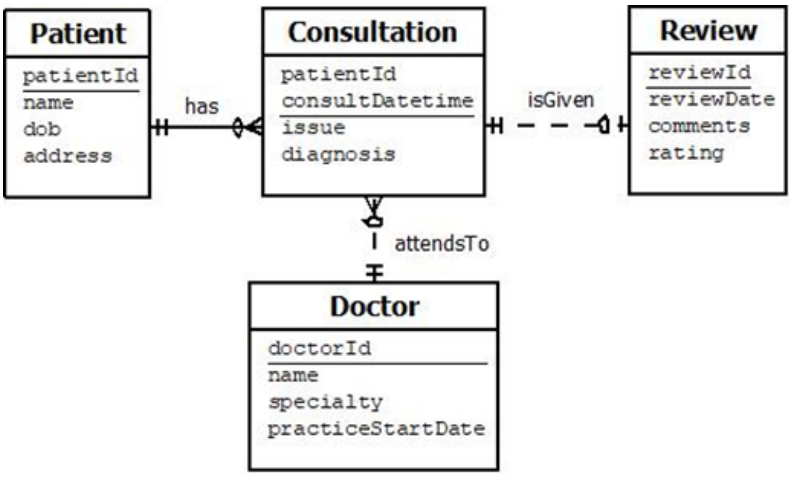

Figure Q3(a)(i)

Show the tables in the logical model in the form:
TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key,
- indicate (FK) against the foreign keys, if any, including integrity constraints in the form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3a(ii)

Refer to the description of the columns for the column definition, and then apply the appropriate SQL statements to **implement ONLY** the **Review table** in Figure Q3(a)(i).

All columns are required except for comments.

patientId	Exactly 5 characters in length, starts and ends with a letter with digits in between,
name (for patient)	At most 60 characters in length
dob	Datetime and value must not be later than the current datetime
address	At most 100 characters in length
consultDatetime	Datetime
issue	At most 1024 characters in length
diagnosis	At most 1024 characters in length
doctorId	Exactly 4 characters in length, starts and ends with a letter with digits in between,
name (for Doctor)	At most 60 characters in length
specialty	Exactly 10 characters in length, with a value either general, gastro, or cardio.
practiceStartDate	Datetime
reviewId	An integer, auto generated with starting value 1 and increment 1
reviewDate	Datetime with default current date and time.
comments	At most 1024 characters in length
rating	A number from 1 to 5

Constraints for the relationships in Figure Q3(a)(i):

has	delete and update are disallowed
attendsTo	delete and update are disallowed
isGiven	delete and update cascaded

(7 marks)

Question 3b

This question is based on the tables derived from the ERD in Figure Q3(a)(i).

Write a trigger to disallow patient from submitting a review for a consultation that has not taken place yet. There is **NO** need to display outcome of the operation (successful or unsuccessful submission).
State any assumptions you made in implementing the trigger.

(6 marks)

Question 3c

For this question part, you are allowed to use inline views in your query formulation, but you must NOT create views using the CREATE VIEW statement.
Use the database tables in Figure Q3(c).

ArtistID	LastName	FirstName	Nationality	DateOfBirth	DateDeceased
1	Miro	Joan	Spanish	1893	1983
2	Kandinsky	Wassily	Russian	1866	1944
3	Klee	Paul	German	1879	1940
4	Matisse	Henri	French	1869	1954
5	Chagall	Marc	French	1887	1985
11	Sargent	John Singer	United States	1856	1925
17	Tobey	Mark	United States	1890	1976
18	Horiuchi	Paul	United States	1906	1999
19	Graves	Morris	United States	1920	2001

Artist table

WorkID	Title	Copy	Medium	Description	ArtistID
570	Untitled Number 1	Unique	Monotype with tempera	4.3 x 6.1 in. Signed	17
571	Yellow Covers Blue	Unique	Oil and collage	71 x 78 in. - Signed	18
578	Mid-Century Hibernation	362/500	High Quality Limited Print	Northwest School Expressionist style	19
580	Forms in Progress I	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
581	Forms in Progress II	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
585	The Fiddler	252/1000	High Quality Limited Print	Shows Belarusian folk-life themes and symbology	5
586	Spanish Dancer	588/750	High Quality Limited Print	American Realist style - From work in Spain	11
587	Broadway Boggie	433/500	High Quality Limited Print	Northwest School Abstract Expressionist style	17
588	Universal Field	114/500	High Quality Limited Print	Northwest School Abstract Expressionist style	17
589	Color Floating in Time	487/500	High Quality Limited Print	Northwest School Abstract Expressionist style	18
590	Blue Interior	Unique	Tempera on card	43.9 x 28 in.	17
593	Surf and Bird	Unique	Gouache	26.5 x 29.75 in. - Signed	19
594	Surf and Bird	362/500	High Quality Limited Print	Northwest School Expressionist style	19
595	Surf and Bird	365/500	High Quality Limited Print	Northwest School Expressionist style	19
596	Surf and Bird	366/500	High Quality Limited Print	Northwest School Expressionist style	19

Work table

TransactionID	DateAcquired	AcquisitionPrice	AskingPrice	DateSold	SalesPrice	CustomerID	WorkID
161	2013-06-28	7500.00	15000.00	2013-09-29	13750.00	1033	570
171	2013-08-23	35000.00	60000.00	2013-09-29	55000.00	1000	571
181	2013-10-11	250.00	500.00	NULL	NULL	NULL	578
201	2014-02-28	2000.00	3500.00	2014-04-26	3250.00	1040	580
202	2014-02-28	2000.00	3500.00	2014-04-26	3250.00	1040	581
225	2014-06-08	125.00	250.00	2014-09-27	225.00	1051	585
226	2014-06-08	200.00	400.00	NULL	NULL	NULL	586
227	2014-06-08	250.00	500.00	2014-07-11	475.00	1051	587
228	2014-06-08	250.00	500.00	NULL	NULL	NULL	588
229	2014-06-08	250.00	500.00	NULL	NULL	NULL	589

241	2014-08-29	2500.00	5000.00	2014-09-27	4750.00	1015	590
251	2014-10-25	25000.00	50000.00	NULL	NULL	NULL	593
252	2014-10-27	250.00	500.00	NULL	NULL	NULL	594
253	2014-10-27	250.00	500.00	NULL	NULL	NULL	595
254	2014-10-27	250.00	500.00	NULL	NULL	NULL	596

Trans table

Note: For rows in Trans table when customerID is NULL, it means that the art works have not been sold off yet.

Figure Q3c

Formulate SQL queries to perform the following tasks:

Question 3c(i)

List the works where the copy is unique and the description contains the word **Signed**. List in ascending order of workID.

WorkID	Title	Copy	Medium	Description	ArtistID
570	Untitled Number 1	Unique	Monotype with tempera	4.3 x 6.1 in. Signed	17
571	Yellow Covers Blue	Unique	Oil and collage	71 x 78 in. - Signed	18
580	Forms in Progress I	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
581	Forms in Progress II	Unique	Color aquatint	19.3 x 24.4 in. - Signed	17
593	Surf and Bird	Unique	Gouache	26.5 x 29.75 in. - Signed	19

(7 marks)

Question 3c(ii)

List the details of work pieces that are currently not sold yet. Rename the column lastName as ArtistLastName.

transactionId	dateAcquired	acquisitionPrice	title	ArtistLastName	firstName
181	2013-10-11	250.00	Mid-Century Hibernation	Graves	Morris
226	2014-06-08	200.00	Spanish Dancer	Sargent	John Singer
228	2014-06-08	250.00	Universal Field	Tobey	Mark
229	2014-06-08	250.00	Color Floating in Time	Horiuchi	Paul
251	2014-10-25	25000.00	Surf and Bird	Graves	Morris
252	2014-10-27	250.00	Surf and Bird	Graves	Morris
253	2014-10-27	250.00	Surf and Bird	Graves	Morris
254	2014-10-27	250.00	Surf and Bird	Graves	Morris

(8 marks)

Question 4

Question 4a

Demonstrate the basic techniques of concurrency control and their application.

Consider the tables in Figure Q4a:

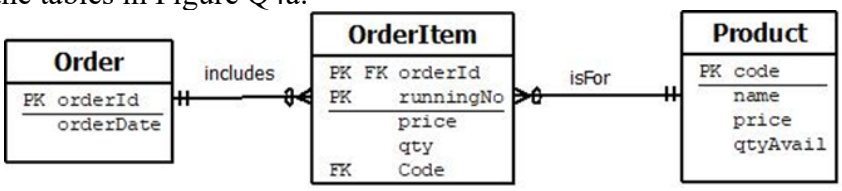

Figure Q4a

Discuss the lost update problem that can arise when a wrong isolation level is used when working on the tables. You should explain the work that concurrent transactions are attempting to do, state the isolation level that leads to the problem discussed as well as to explain how the problem can arise. Identify the table(s) and column(s) involved as well as provide a clear description of interleaving actions that cause the lost update to occur.

(6 marks)

Question 4b

Describe **ONE (1)** permission that should be given to registered customers using the database for Question 4a, based on the tables in Figure Q4(a). Describe **ONE (1)** permission that should not be given to registered customers. Specify the permissions in terms of these actions: select, delete, insert and update on user tables in Figure Q4(a).

(4 marks)

Question 4c

Given Figure 4(c), explain what the database recovery procedure will do for the transactions T1 and T2.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	9:01:00	Start				
2	T1	1	3	9:01:36	Update	Table1	2	2, 12, …	2, 16, …
3	T1	2	6	9:02:55	Update	Table2	14	14, 2, 7, …	14, 2, 5, …
4	T2	0	5	9:03:01	Start				
5	T2	4	7	9:03:24	Update	Table3	15	15, 3, …	15,5, …
6	T1	3	0	9:04:00	Rollback				
Checkpoint									
7	T2	5	0	9:15:42	Rollback				
Crash!									

Figure 4(c)

(10 marks)

----- END OF PAPER -----